Kai Pietschner

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Canberra: Whitlam 2611 ACT | **Sydney:** Ingleside 2101 NSW

Summary

I am a motivated **penultimate-year Renewable Energy Engineering student** at the **Australian National University** with strong academic performance and experience in engineering projects, communication. I bring technical skills in programming, prototyping, modelling, and data analysis and show great leadership and teamwork. Driven by a passion for renewable energy technology, I am seeking opportunities to practically apply my engineering knowledge, to grow professionally and to contribute to meaningful, real-world engineering projects.

Education

Bachelor of Engineering Honours (Renewable Energy Major, Humanitarian Engineering Minor, Science Communication Minor) at **The Australian National University** (Feb. 2023 – Present)

- **Relevant Coursework:** Engineering Design 1/2/3/4A/4B, Physics 1, Programming for Scientists, Calculus and Linear Algebra 1, Mechanics 1/2, Electronics 1/2, Sustainable Development, Culture and Development, Science Politics, Wind Energy, Energy Resources, Engineering Thermodynamics, Global Crisis Science Communication, Power Electronics and Systems, Engineering Capstone Design Project

International Baccalaureate (IB) Diploma at **The German International School Sydney (GISS)** (Jan. 2016 – Nov. 2022)

- **Courses: High Level Physics**, **High Level Mathematics**: Applications and Interpretations, High Level English, Standard Level Economics, **Standard Level Chemistry**, Standard Level German B: Language Acquisition
- **IB Score of 31** (ATAR Conversion of 85)

Relevant Engineering Projects

Engineering Projects

- **Capstone Project (ENGN4300)** (Semesters 1 & 2 2026): Start-to-finish development of a proof-of-concept product for project hosts, application of all engineering knowledge, engaging stakeholders, and professional documentation
- **Photovoltaics (ENGN4524) Major Project** (Semester 1 2026): Resource and economic model, and case-study creation for feasibility of urban solar-battery installation, 18-page report
- **Wind Energy (ENGN4548) Major Project** (Semester 2 2025): Site analysis for energy productivity, wind resource modelling, cost analysis, feasibility analysis, environmental analysis, 14-page report, and 8-minute presentation
- **Energy Resources (ENGN3516) Major Project** (Semester 2 2025): Challenges and opportunities in wind and PV integration 18-page group report, with NEM stakeholder
- **Many more** (can be found on my portfolio website)

Work Experience

Sales Assistant/Specialist at **Jaycar Electronics** (2022 – Current):

- Knowledgeable and specific advice for solving electronics problems for trades people and hobbyists

Work Experience at **Capgmini** (2020):

- Shadowed events and meetings group for IT systems consulting

Front Office Admin & Coach at **Evolve Tennis Academy** (2020 – 2022):

- Managed bookings, customer inquiries, payment services and coached as part of a team

Key Skills

- **Programming and Software:** OpenWind, LTSpice, ANSYS, Python, Arduino IDE, MATLAB, Microsoft Office Suite (Word, Excel, PowerPoint, Outlook), “Digital” circuit simulator
- **Languages:** German B2-A1 (DSD 1)
- **Engineering and Laboratory Skills:** Systems Engineering Project Management Documentation, Solar/Wind Resource Analysis, Statistical Data Analysis, Computational Problem-Solving Techniques, Circuit Analysis and Design, Material Testing and Failure Analysis, Prototyping and Testing, 3D modelling.

References available upon request.